CLUSTERING

CLUSTER VALIDITY – PART III

Fabio Stella

Associate Professor

c/o Department of Informatics, Systems and Communication

University of Milano-Bicocca

Cluster Validity

The following concept will be introduced:

✓ **Validity Paradigm**

IDENTIFY THE CLUSTERING STRUCTURE AND VALIDATION TYPE

EXTERNAL OR INTERNAL?

Both **EXTERNAL AND INTERNAL CRITERIA** are closely related to statistical methods and hypothesis tests.

The **VALIDITY PARADIGM**, for a clustering structure, is based on the following **NULL HYPOTHESIS**

"**THERE IS NO STRUCTURE ON THE DATASET**"

CLUSTER VALIDITY: THE VALIDITY PARADIGM

1

IDENTIFY THE CLUSTERING STRUCTURE AND VALIDATION TYPE

DETERMINE A VALIDATION INDEX

WHICH SPECIFIC VALIDATION INDEX?

IDENTIFY THE CLUSTERING STRUCTURE AND VALIDATION TYPE

DETERMINE A VALIDATION INDEX

DEFINE A NULL HYPOTHESIS OF NO STRUCTURE

Three commonly used null hypotheses

- **RANDOM POSITION HYPOTHESIS**, all the locations of the **m** data points in some specific region of a n-dimensional space are equally likely ← **FOR RATIO DATA**
- **RANDOM GRAPH HYPOTHESIS**, all [**m**×**m**] rank order proximity matrices are equally likely ← **FOR ORDINAL PROXIMITIES BETWEEN PAIRS OF DATA OBJECTS**
- **RANDOM LABEL HYPOTHESIS**, all permutations of the labels on **m** data points are equally likely ← **FOR ALL DATA TYPES**

Identify the clustering structure and validation type

↓

Determine a validation index

↓

Define a null hypothesis of no structure

↓

Establish the baseline distribution under the null hypothesis

Monte Carlo analysis

and Bootstrapping

Cluster Validity: The Validity Paradigm

1

Identify the clustering structure and validation type

↓

Determine a validation index

↓

Define a null hypothesis of no structure

↓

Establish the baseline distribution under the null hypothesis

↓

Calculate the index

CLUSTER VALIDITY: THE VALIDITY PARADIGM

IDENTIFY THE CLUSTERING STRUCTURE AND VALIDATION TYPE

↓

DETERMINE A VALIDATION INDEX

↓

DEFINE A NULL HYPOTHESIS OF NO STRUCTURE

↓

ESTABLISH THE BASELINE DISTRIBUTION UNDER THE NULL HYPOTHESIS

↓

CALCULATE THE INDEX

↓

TEST THE HYPOTHESIS OF NO STRUCTURE

You hope to reject the null hypothesis that there is no structure in the clustering you have developed